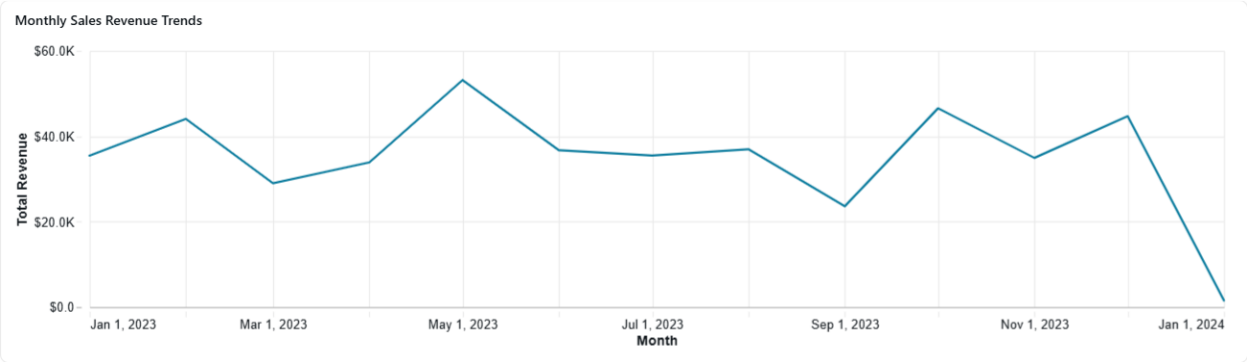
Comprehensive Retail Sales Analytics Report

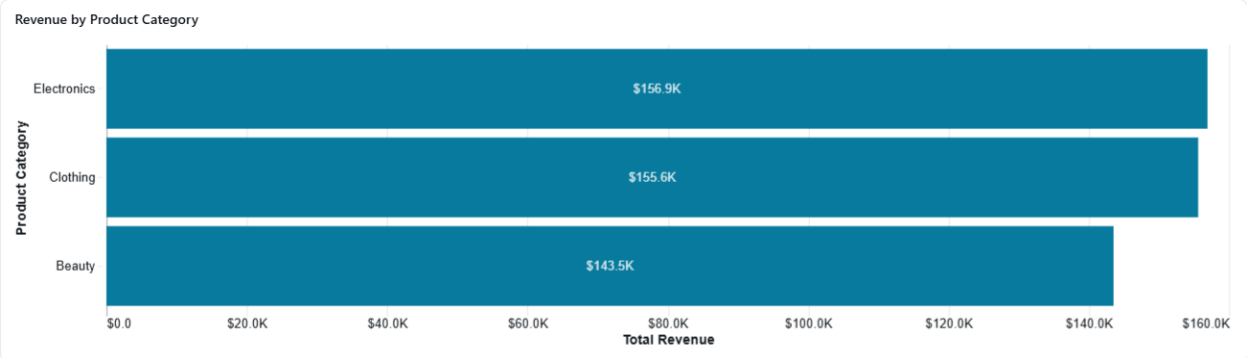
Monthly Sales Revenue Trends

Sales revenue shows significant **monthly fluctuations** throughout 2023, with **May 2023 generating the highest revenue at $53,150** and September 2023 recording the lowest at $23,620. The data shows strong performance in February ($44,060), October ($46,580), and December ($44,690), suggesting potential seasonal patterns with peaks in spring and year-end periods.



Product Category Revenue Performance

Electronics leads in total revenue with $156,905, followed closely by Clothing at $155,580 and Beauty at $143,515. The revenue distribution is remarkably balanced across all three categories, with Electronics holding only a 1.7% advantage over Clothing and a 9.3% advantage over Beauty.

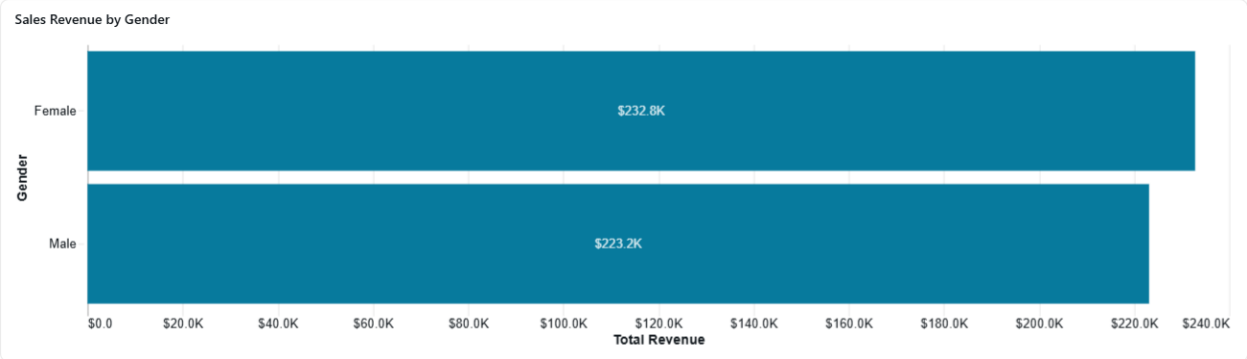


However, when examining **quantity sold, Clothing leads with 894 units**, followed by Electronics (849 units) and Beauty (771 units). This indicates that while Clothing sells more units, Electronics generates higher revenue per unit, suggesting a premium pricing strategy.

Gender-Based Sales Comparison

Female customers generate slightly higher revenue at $232,840 compared to male customers at $223,160, representing a 4.3% difference. Female customers also completed more transactions

(510 vs 490), though the average transaction values are nearly identical ($456.55 for females vs $455.43 for males).



Product Category Preferences by Gender

Gender preferences show distinct patterns: **Female customers prefer Clothing** ($81,275 revenue), while **male customers favor Electronics** ($80,170 revenue). Beauty products generate the lowest revenue from male customers ($68,685), suggesting a stronger female customer base for this category.



Customer Spend by Age Group

Younger customers (18-24) have the highest average spend at $501.01, followed by 25-34 age group at $478.28. Average spend decreases with age, with the 55+ group showing the lowest average at $417.55. However, the 45-54 age group represents the largest customer base with 225 unique customers and generates $97,235 in total revenue.



Top 10 Customers

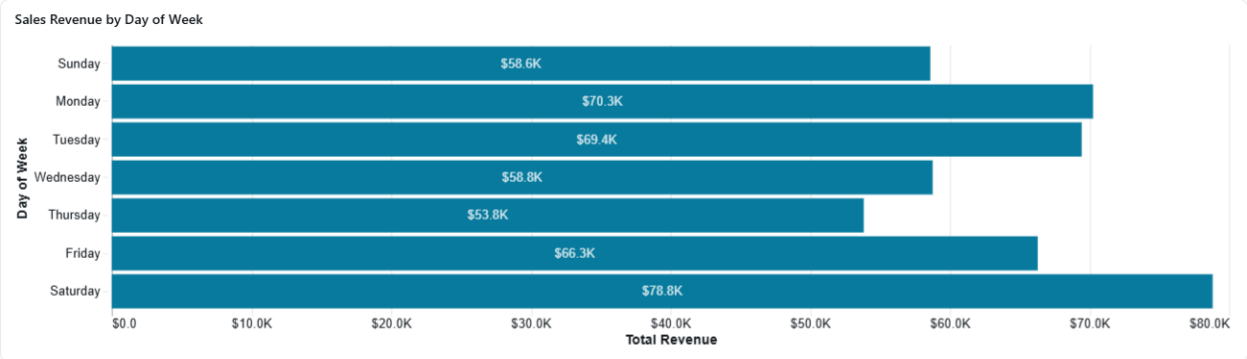
The top 10 customers each spent exactly **$2,000 in a single transaction**, representing high-value purchases. This suggests these are likely premium product purchases, possibly in the Electronics category given its higher average transaction value.

Top 10 Customers by Total Spend

Customer ID	Total Spend	Transactions	Avg Transaction Value
CUST109	$2,000	1	$2,000.00
CUST089	$2,000	1	$2,000.00
CUST074	$2,000	1	$2,000.00
CUST124	$2,000	1	$2,000.00
CUST072	$2,000	1	$2,000.00
CUST093	$2,000	1	$2,000.00
CUST015	$2,000	1	$2,000.00
CUST065	$2,000	1	$2,000.00
CUST118	$2,000	1	$2,000.00
CUST139	$2,000	1	$2,000.00

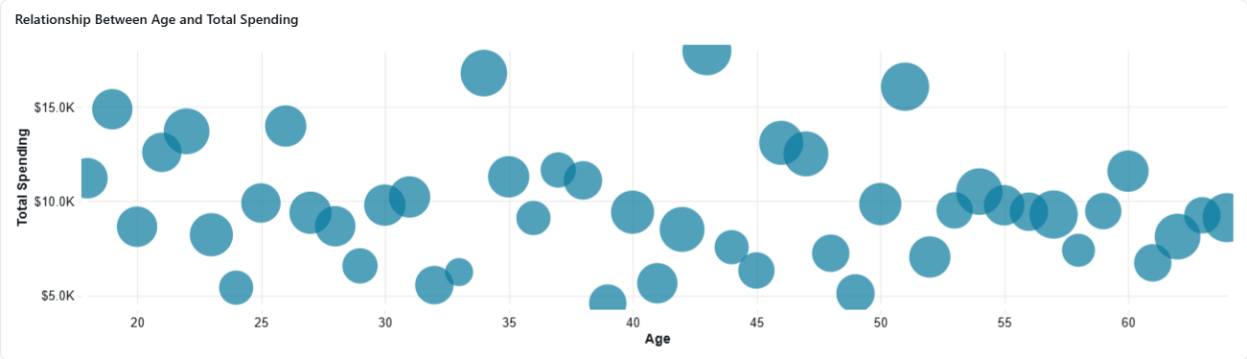
Day of Week Performance

Saturday generates the highest sales revenue at $78,815, followed by Monday ($70,250) and Tuesday ($69,440). Thursday shows the lowest revenue at $53,835. Weekend shopping (Saturday-Sunday combined: $137,415) significantly outperforms mid-week periods, with Saturday also showing the highest average transaction value at $525.43.



Age and Spending Relationship

The scatter plot reveals **no strong linear correlation between age and total spending**. Spending patterns are distributed across all age groups, with notable peaks at ages 19, 34, 43, and 51. The bubble size (representing transaction count) shows that spending is more influenced by transaction frequency than age alone.



Strategic Recommendation: Prioritize Electronics

Management should prioritize Electronics for future growth based on the following evidence:

1.
Highest Revenue Generation: Electronics leads with $156,905 in total revenue, representing 34.4% of total sales
2.
Premium Pricing Power: Electronics achieves the highest average transaction value at $458.79, compared to Clothing ($443.25) and Beauty ($467.48), while selling fewer units (849 vs 894 for Clothing)
3.
Strong Male Customer Appeal: Electronics generates $80,170 from male customers, the highest revenue from any gender-category combination, providing a balanced customer base alongside female preferences for Clothing

4.
High-Value Customer Attraction: The top 10 customers with $2,000 single transactions likely purchased Electronics products, indicating this category attracts premium buyers
5.
Growth Potential: With 342 unique customers and strong performance across demographics, Electronics has room to expand market share through targeted marketing, especially on high-traffic days like Saturdays and Mondays

Recommended Actions: Increase Electronics inventory for weekend sales, develop targeted campaigns for male customers aged 25-44, and introduce premium product lines to capitalize on the high average transaction value.